

Autonomous Mobile Robots and Programming

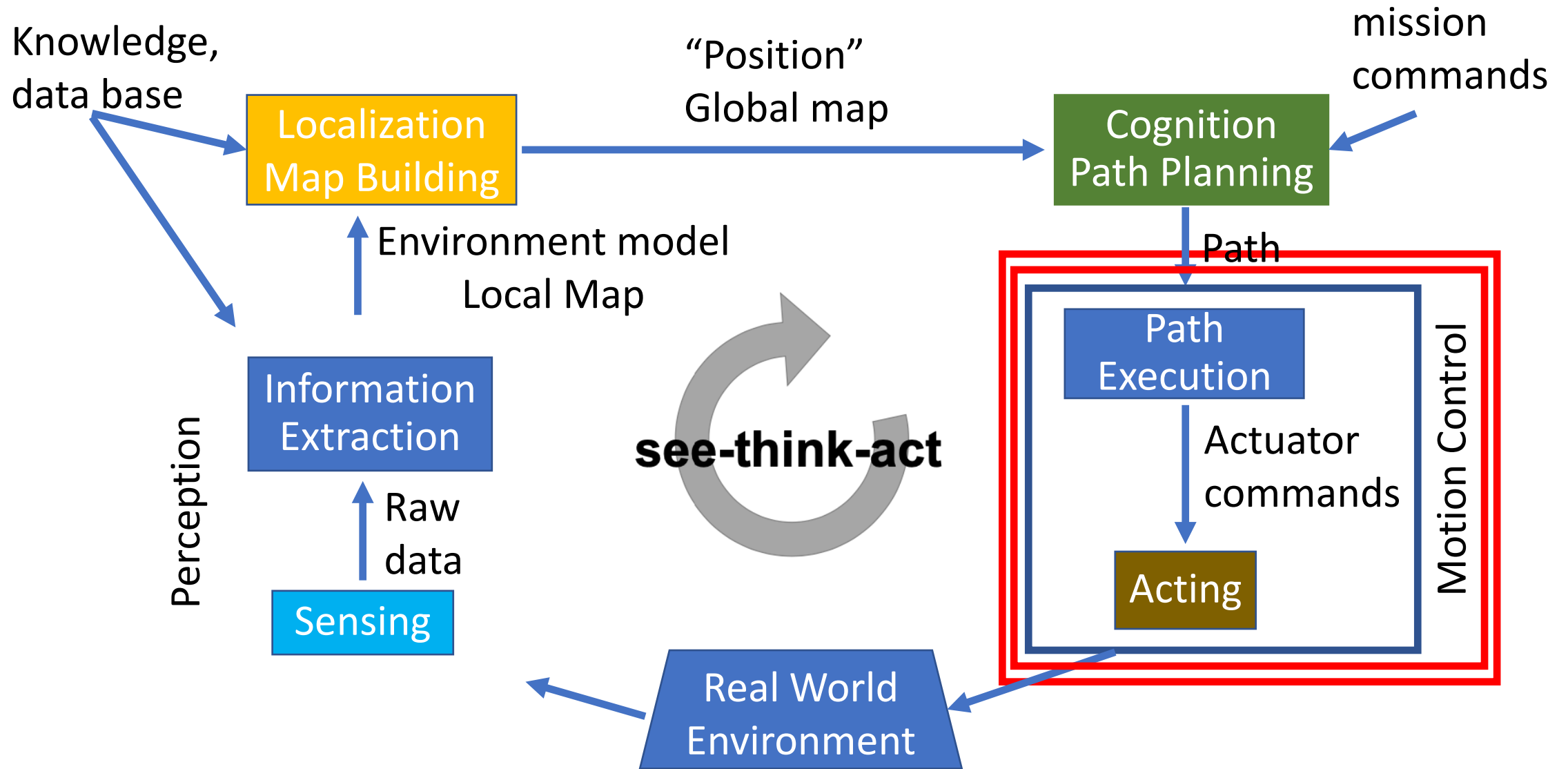
Lecture 2: Locomotion Concepts

Instructor: Dr. Qi (Luki) Lu

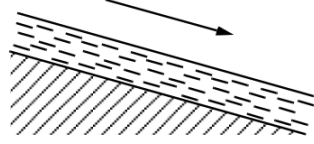
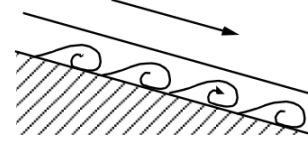
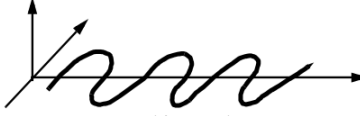
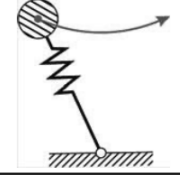
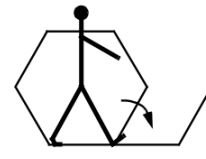
Department of Computer Science

The University of Texas at Rio Grande Valley

The see-think-act cycle



Locomotion Concepts: Principles Found in Nature

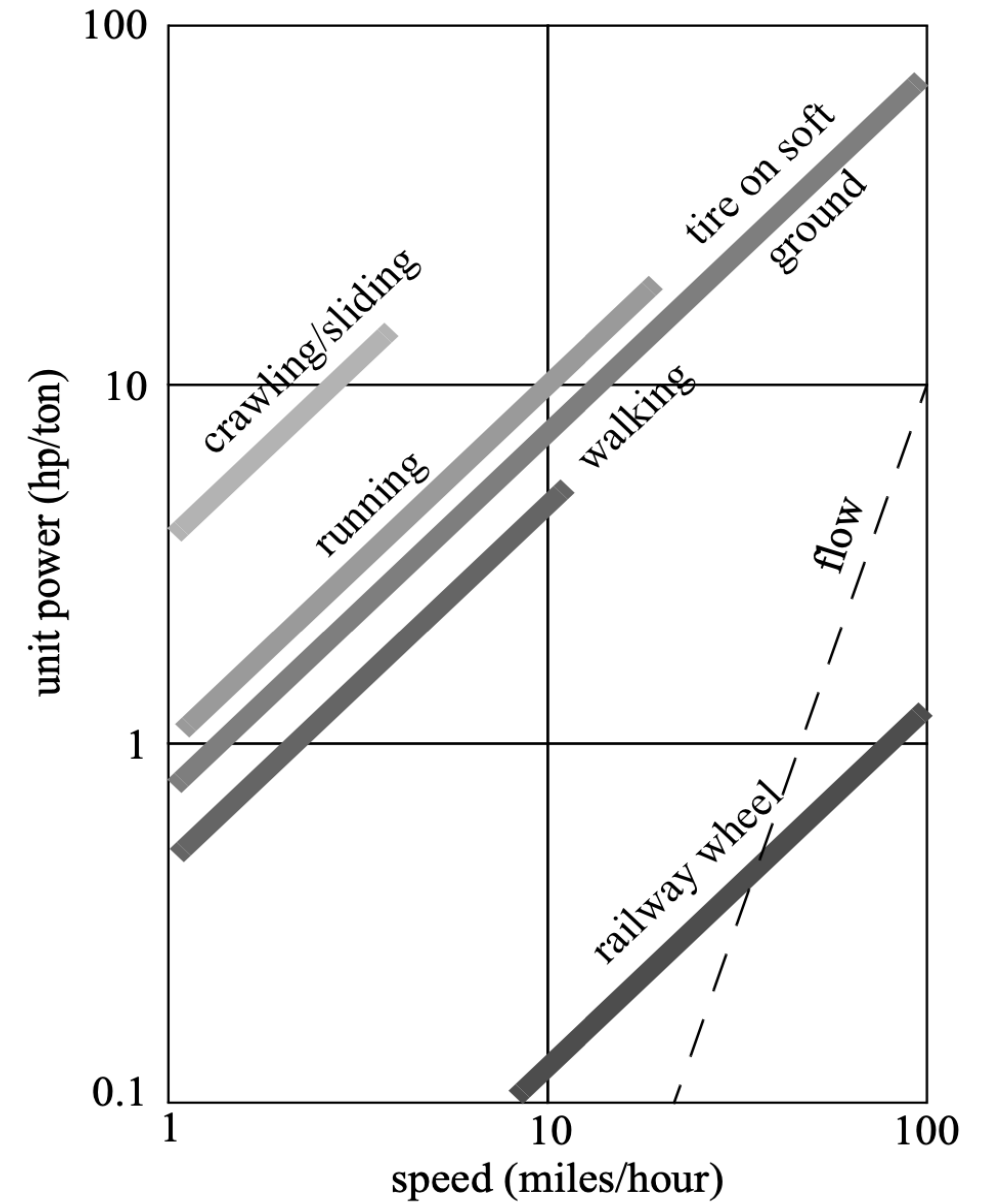
Type of motion	Resistance to motion	Basic kinematics of motion
Flow in a Channel 	Hydrodynamic forces	Eddies 
Crawl 	Friction forces	Longitudinal vibration 
Sliding 	Friction forces	Transverse vibration 
Running 	Loss of kinetic energy	Periodic bouncing on a spring 
Walking 	Loss of kinetic energy	Rolling of a polygon (see figure 2.2) 

Locomotion Concepts

- Nature evolved a multitude of locomotion concepts
 - Adaptation to environmental characteristics
 - Adaptation to the perceived environment (e.g., size)
- Concepts found in nature
 - Difficult to imitate technically
 - Do not employ wheels
 - Sometimes imitate wheels (bipedal walking)
 - The smaller living creatures are, the more likely they fly
- Most technical systems today use wheels or caterpillars
 - Legged locomotion is still mostly a research topic

Walking or rolling?

- Number of actuators
- Structural complexity
- Control expense
- Energy efficient
 - terrain (flat ground, soft ground, climbing..)
 - Cost of Transportation
 - $CoT = \frac{E}{m \cdot g \cdot d} = \frac{P}{m \cdot g \cdot v} = C_{mt}$
 - $E = \text{Energy}; P = \text{Power}; m = \text{mass};$
 - $d = \text{distance}; v = \text{speed}$
- Movement of the involved masses
 - walking / running includes up and down movement of COG
 - some extra losses



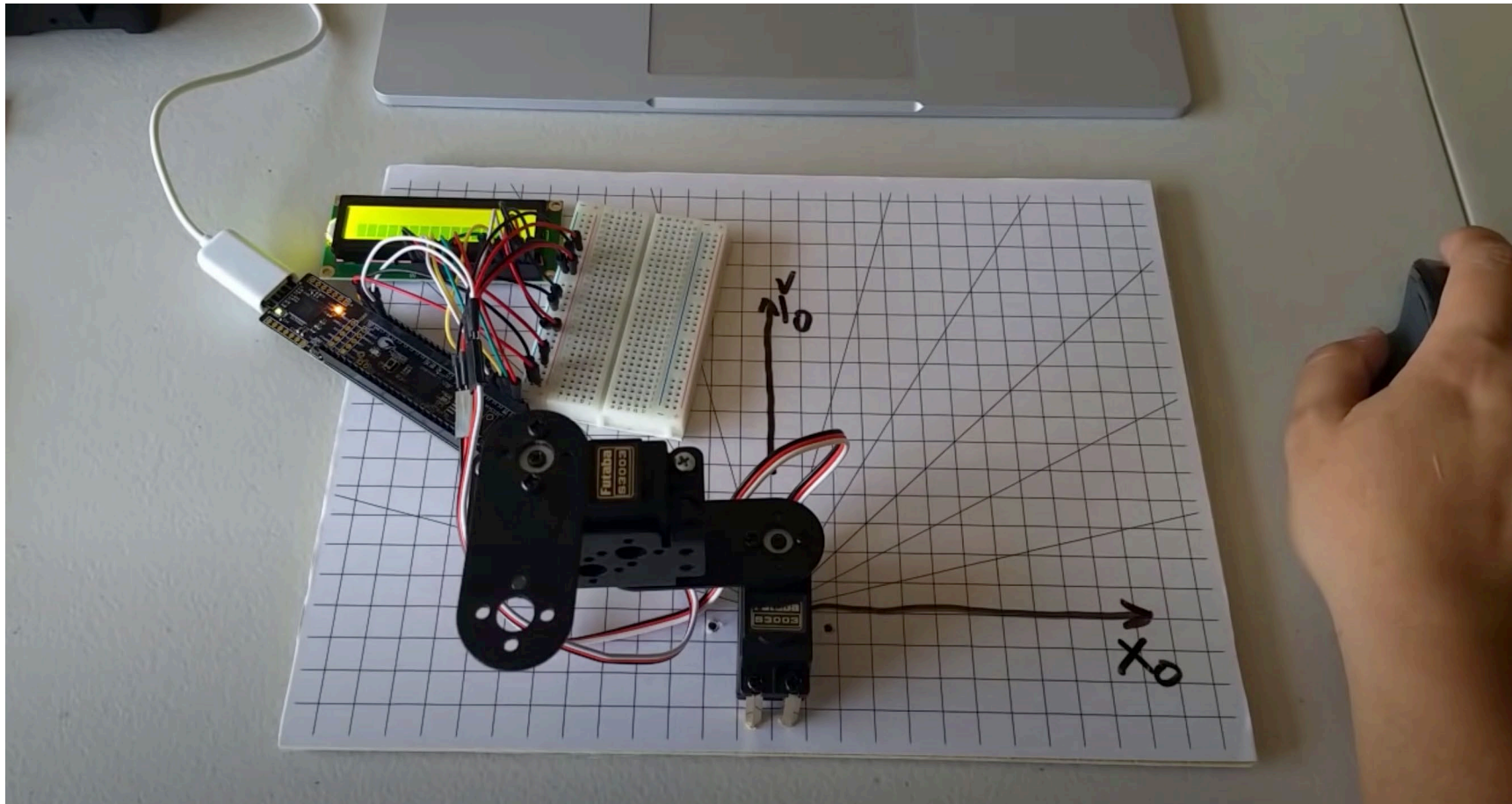
Specific power versus attainable speed of various locomotion mechanisms

Characterization of locomotion concept

- Locomotion
 - physical interaction between the vehicle and its environment.
- Locomotion is concerned with interaction forces, and the mechanisms and actuators that generate them.
- The most important issues in locomotion are:
 - **Stability**
 - number of contact points
 - center of gravity
 - static/dynamic stabilization
 - inclination of terrain
 - **Characteristics of contact**
 - contact point or contact area
 - angle of contact
 - Friction
 - **Type of environment**
 - structure
 - medium (water, air, soft or hard ground)

Motion Control: Kinematics & motion control

- Representing robot position



Motion Control: Kinematics & motion control

- Representing robot position

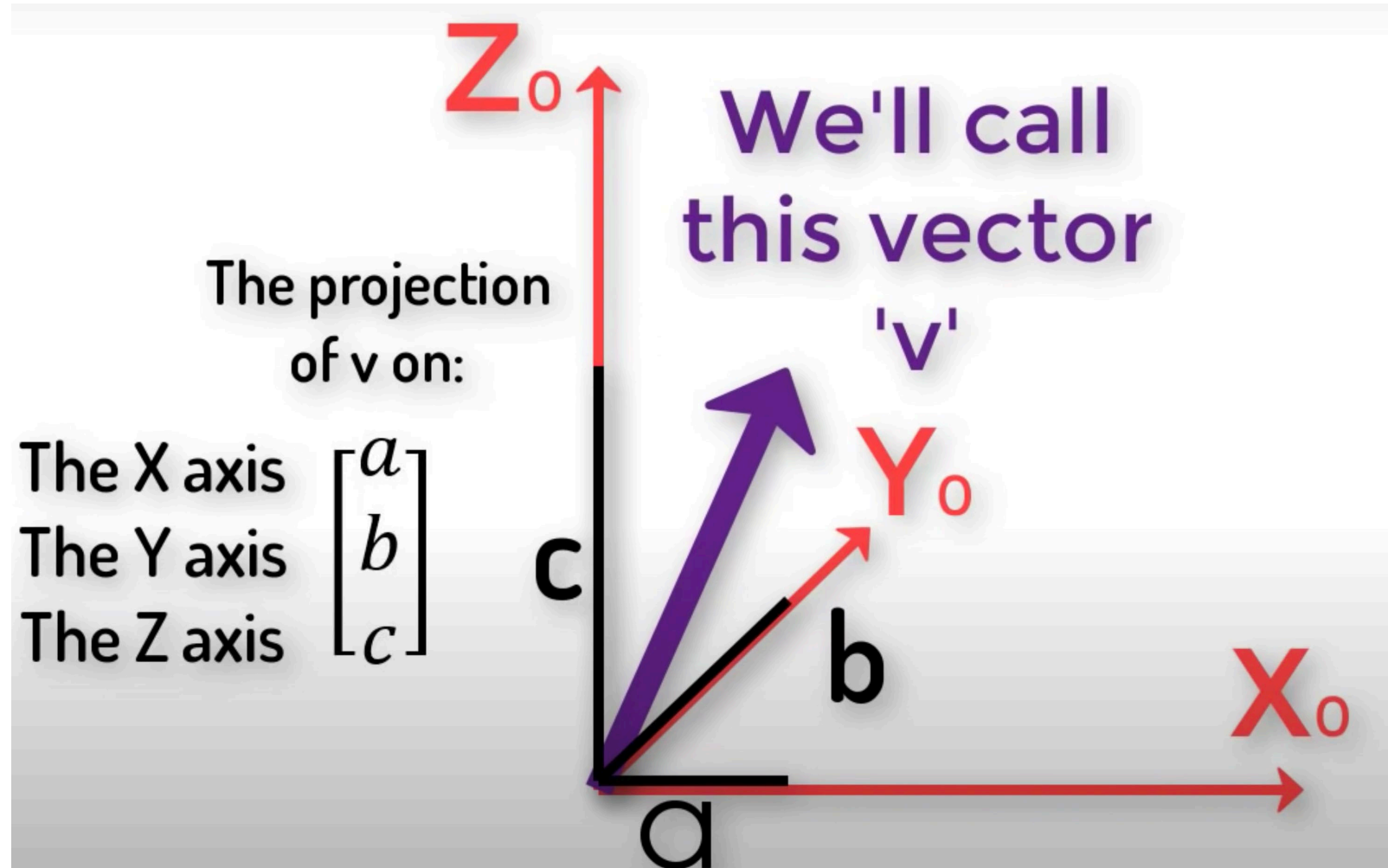


Motion Control: Kinematics & motion control

- Representing robot position
- Represented by
 - **Positions**: Displacement vectors
 - **Rotations**: Rotation matrices

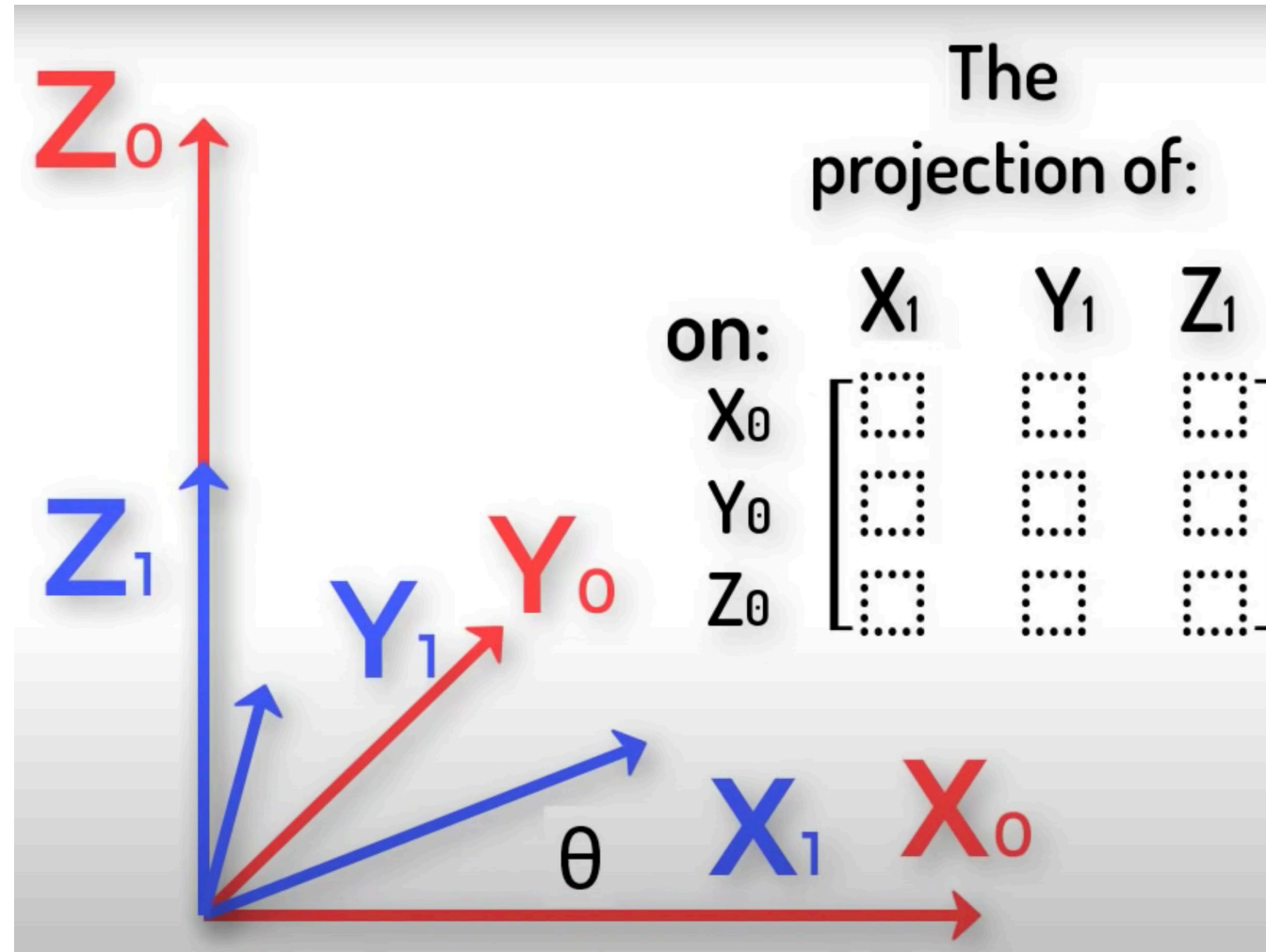
Motion Control: Kinematics & motion control

- Projection



Motion Control: Kinematics & motion control

- Projection



Motion Control: Kinematics & motion control

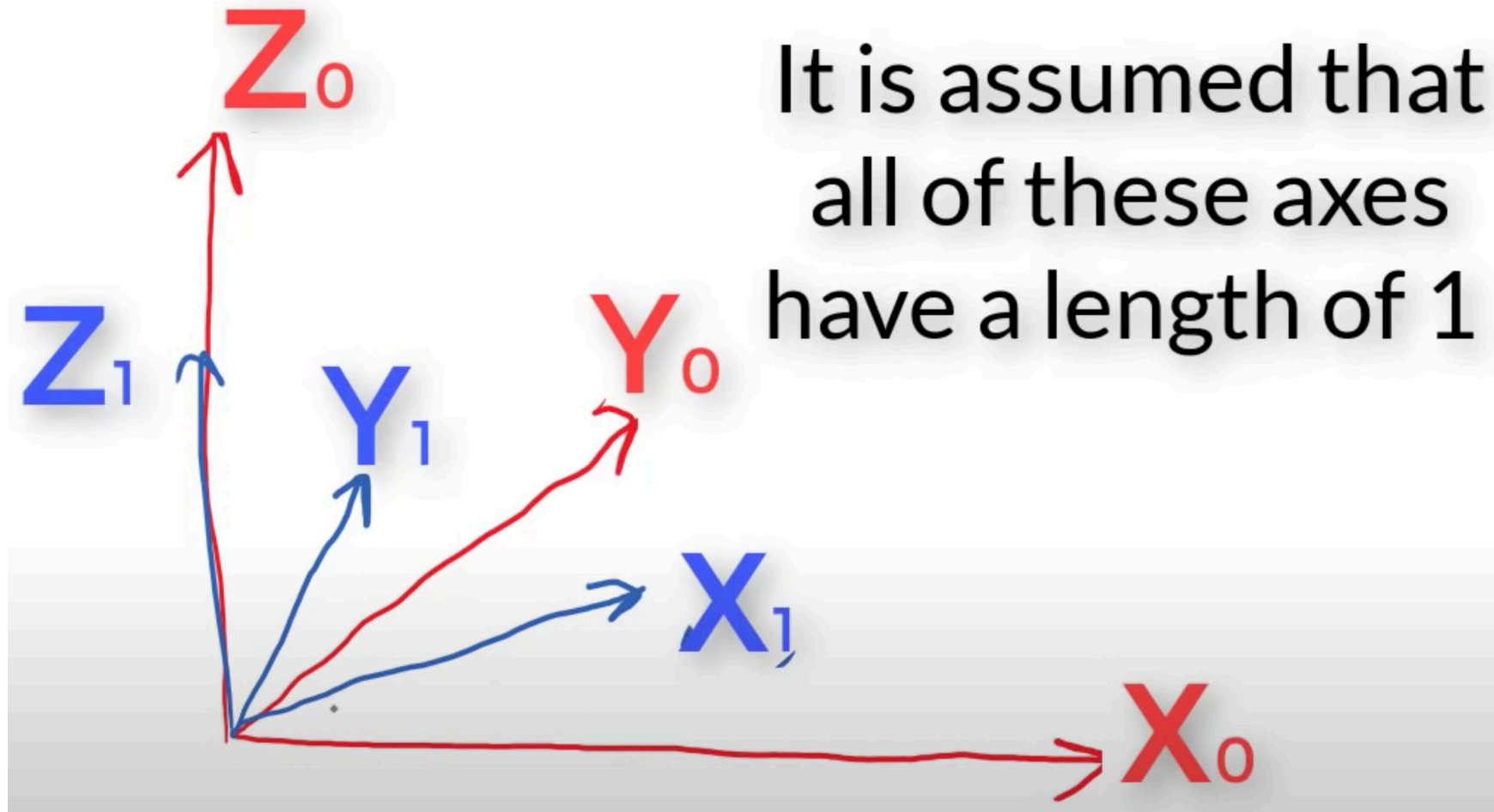
- Rotation Matrix

The
projection of:

$$R_1^0 = \begin{matrix} & \text{on:} & X_1 & Y_1 & Z_1 \\ \begin{matrix} X_0 \\ Y_0 \\ Z_0 \end{matrix} & & \begin{bmatrix} \square & \square & \square \\ \square & \square & \square \\ \square & \square & \square \end{bmatrix} \end{matrix}$$

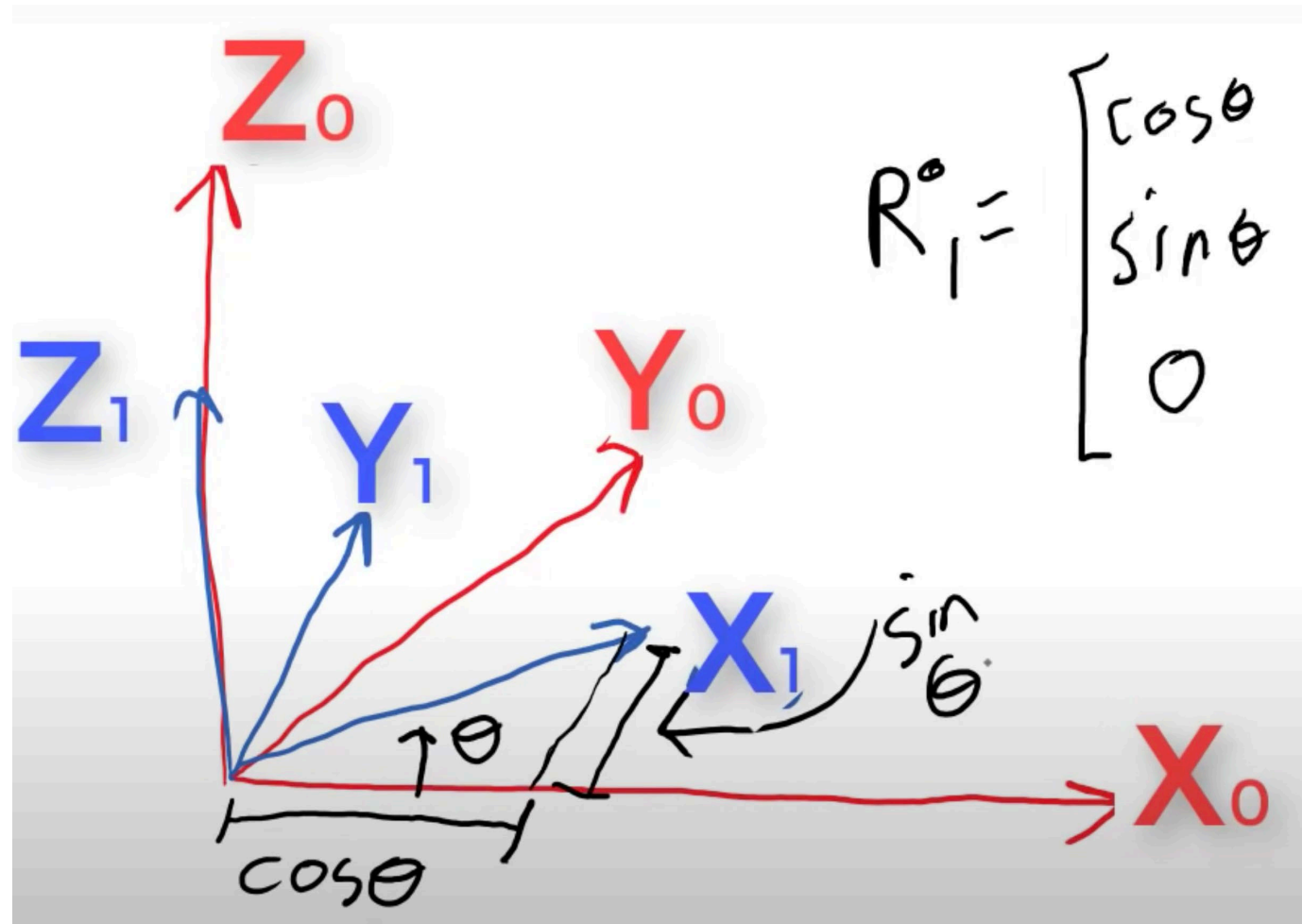
Motion Control: Kinematics & motion control

- Rotation Matrix



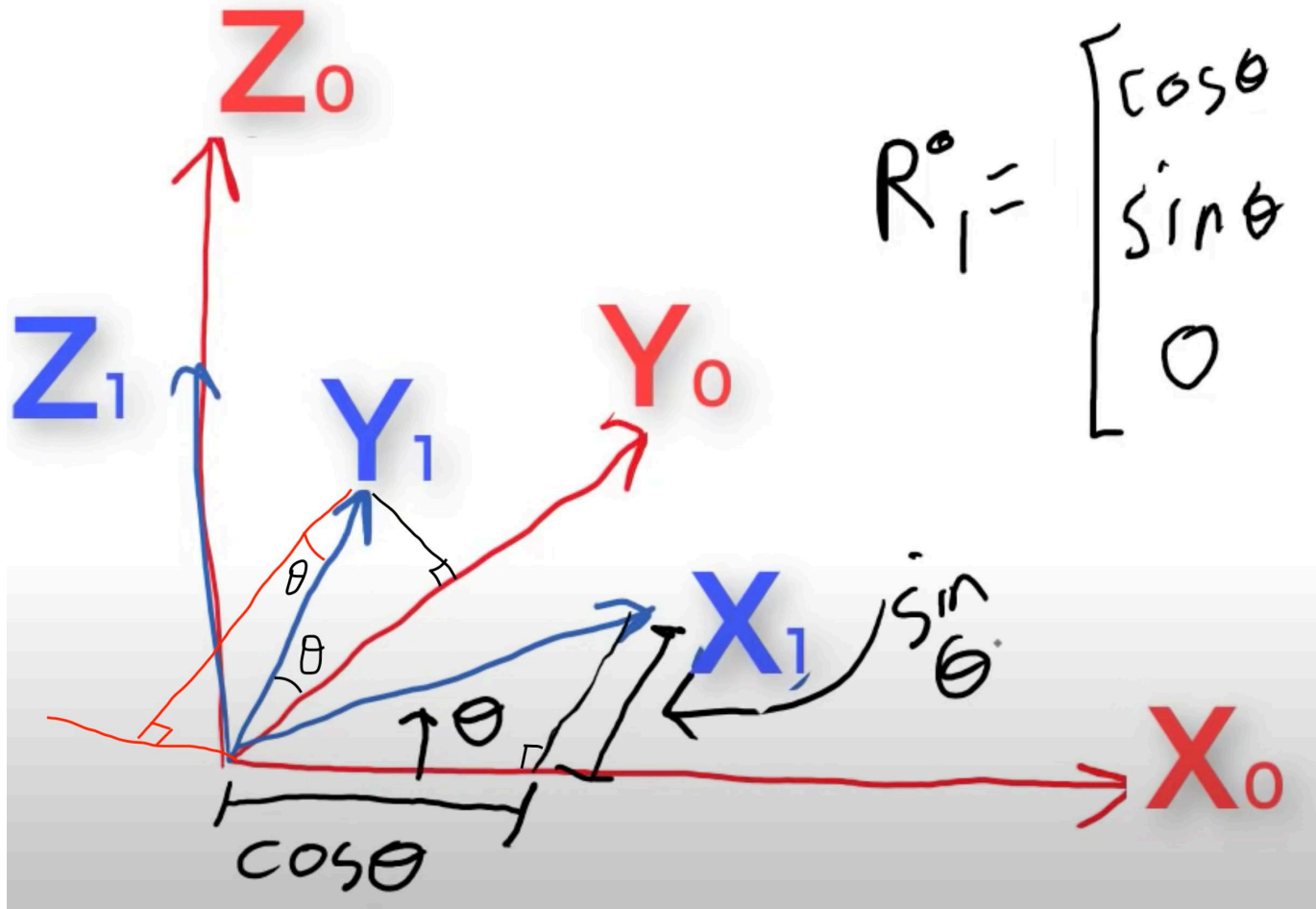
Motion Control: Kinematics & motion control

- Rotation Matrix



Motion Control: Kinematics & motion control

- Rotation Matrix

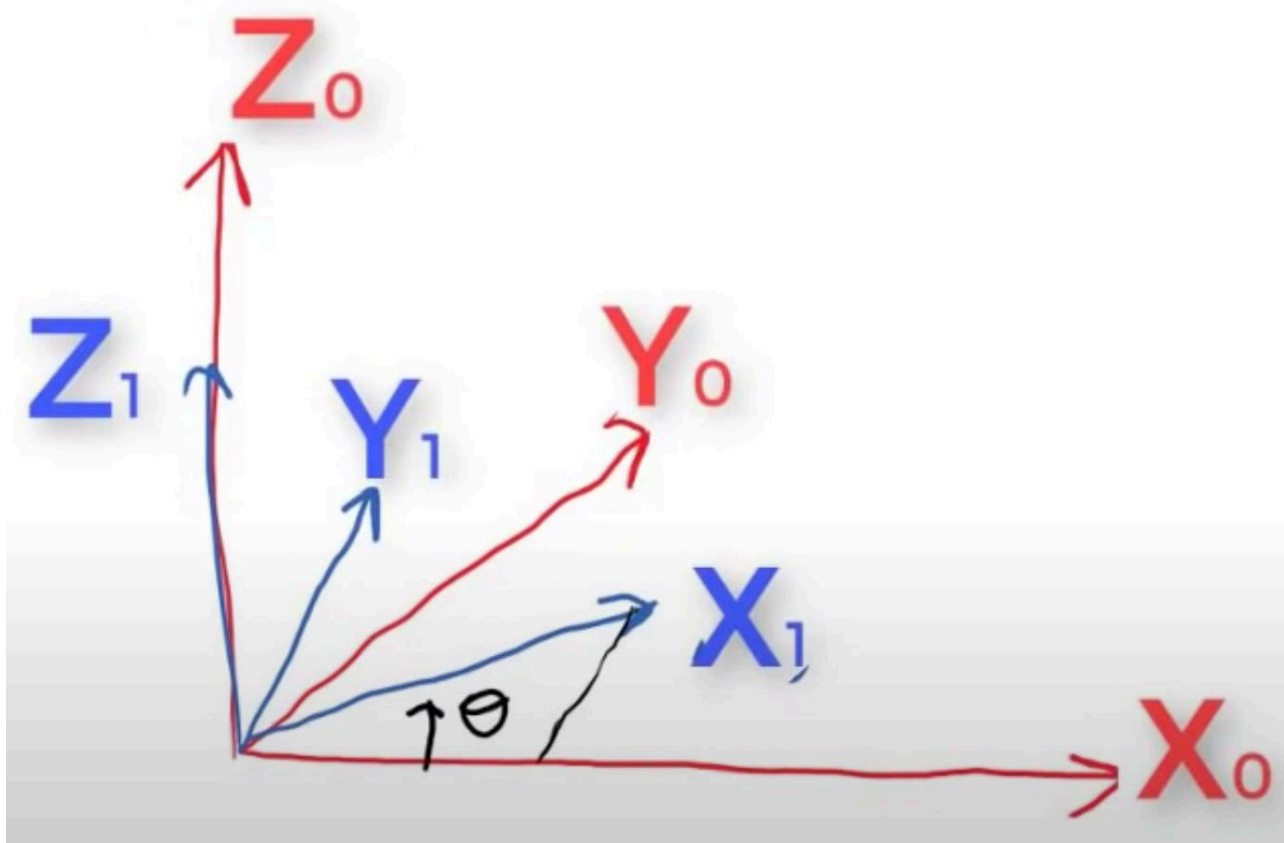


$$R^0_1 = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

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Motion Control: Kinematics & motion control

- Rotation Matrix

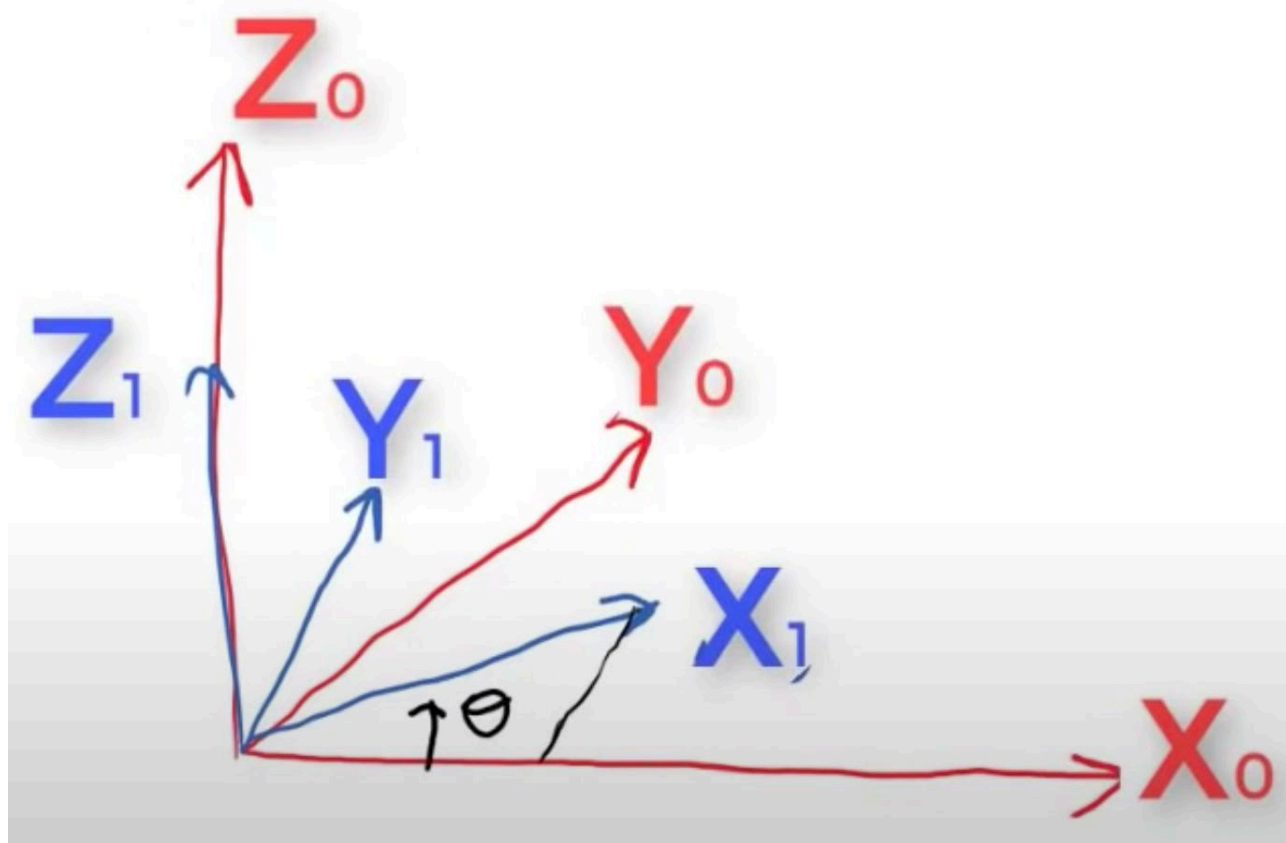


$$R^0_1 = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- The matrix represents the rotation on Z axis with the angle θ

Motion Control: Kinematics & motion control

- Rotation Matrix



$$R_1^0 = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

If the position is $\langle 1, 1, 2 \rangle$ in $\langle X_0, Y_0, Z_0 \rangle$, and $\theta = 45^\circ$, what is the position in $\langle X_1, Y_1, Z_1 \rangle$?

$$\begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ \sqrt{2} \\ 2 \end{bmatrix}$$

Motion Control: Kinematics & motion control

- Rotation Matrix

$$R_X = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}$$

$$R_Y = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}$$

$$R_Z = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Motion Control: Kinematics & motion control

- To rotate the vector $[x \ y \ z]$ counterclockwise through an angle θ around the x – axis to a new position $[x' \ y' \ z']$, perform the matrix multiplication,

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

Motion Control: Kinematics & motion control

- To rotate the vector $[x \ y \ z]$ counterclockwise through an angle θ around the $y - \text{axis}$ to a new position $[x' \ y' \ z']$, perform the matrix multiplication,

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

- To rotate the vector $[x \ y \ z]$ counterclockwise through an angle θ around the $z - \text{axis}$ to a new position $[x' \ y' \ z']$, perform the matrix multiplication,

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

Example 1

- Find the vector $[x' \ y' \ z']$ that results when the vector $[x \ y \ z] = [1 \ 2 \ 3]$ is rotated 90° counterclockwise around x - axis.

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$
$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos 90^\circ & -\sin 90^\circ \\ 0 & \sin 90^\circ & \cos 90^\circ \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

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[https://math.libretexts.org/Bookshelves/Applied Mathematics/Mathematics for Game Developers \(Burzynski\)/04%3A Matrices/4.06%3A Rotation Matrices in 3-Dimensions](https://math.libretexts.org/Bookshelves/Applied_Mathematics/Mathematics_for_Game_Developers_(Burzynski)/04%3A_Matrices/4.06%3A_Rotation_Matrices_in_3-Dimensions)

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix}$$

Example 2

- Find the vector $[x' \ y' \ z']$ that results when the vector $[x \ y \ z] = [1 \ 2 \ 3]$ is rotated 45° counterclockwise around z - axis.

Example 2

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$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$
$$= \begin{bmatrix} \cos 45^\circ & -\sin 45^\circ & 0 \\ \sin 45^\circ & \cos 45^\circ & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

Example 2

- Find the vector $[x' \ y' \ z']$ that results when the vector $[x \ y \ z] = [1 \ 2 \ 3]$ is rotated 45° counterclockwise around z - axis.

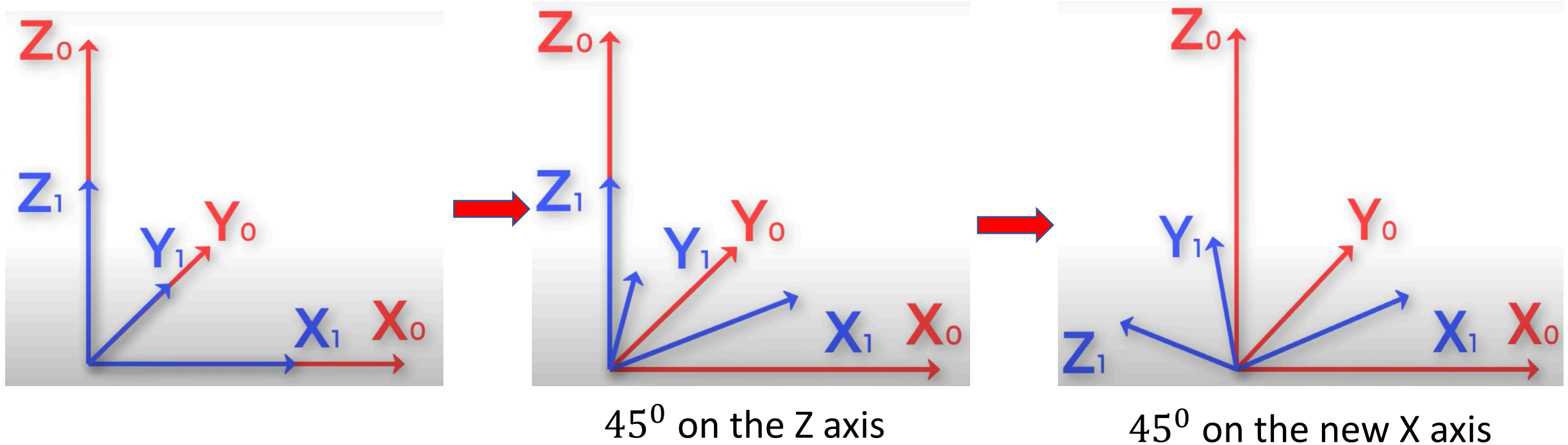
$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$= \begin{bmatrix} \cos 45^\circ & -\sin 45^\circ & 0 \\ \sin 45^\circ & \cos 45^\circ & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} -1/\sqrt{2} \\ 3/\sqrt{2} \\ 3 \end{bmatrix}$$

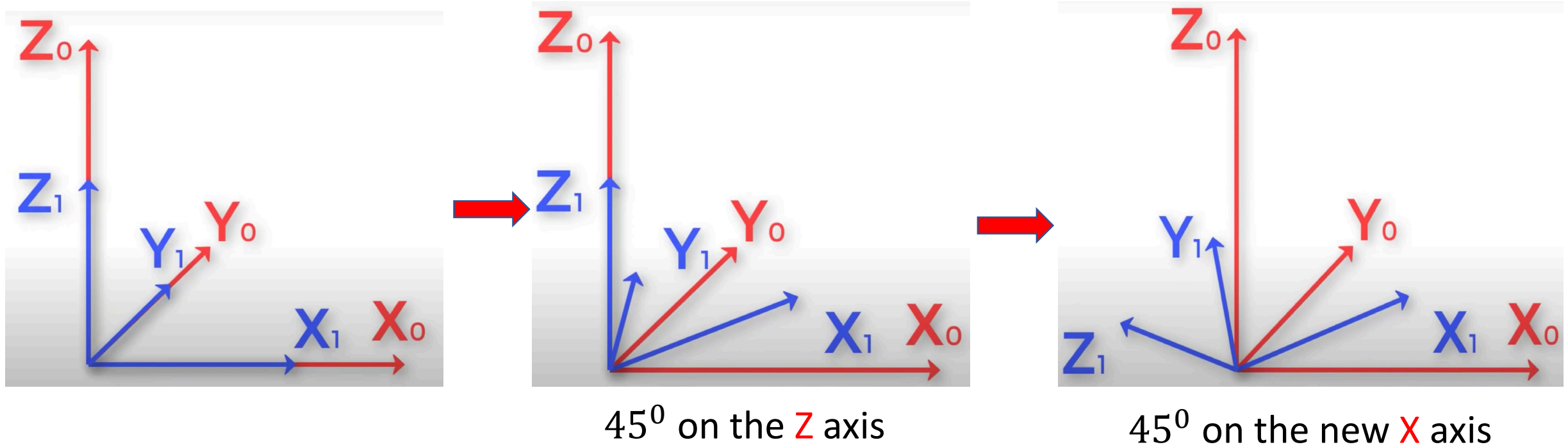
Motion Control: Kinematics & motion control

- Rotation on multiple axes



Motion Control: Kinematics & motion control

- Rotation on multiple axes (**the order matters**)



$$R_{XZ} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$V_{new} = R_{XZ} * V$$

Walking Robots

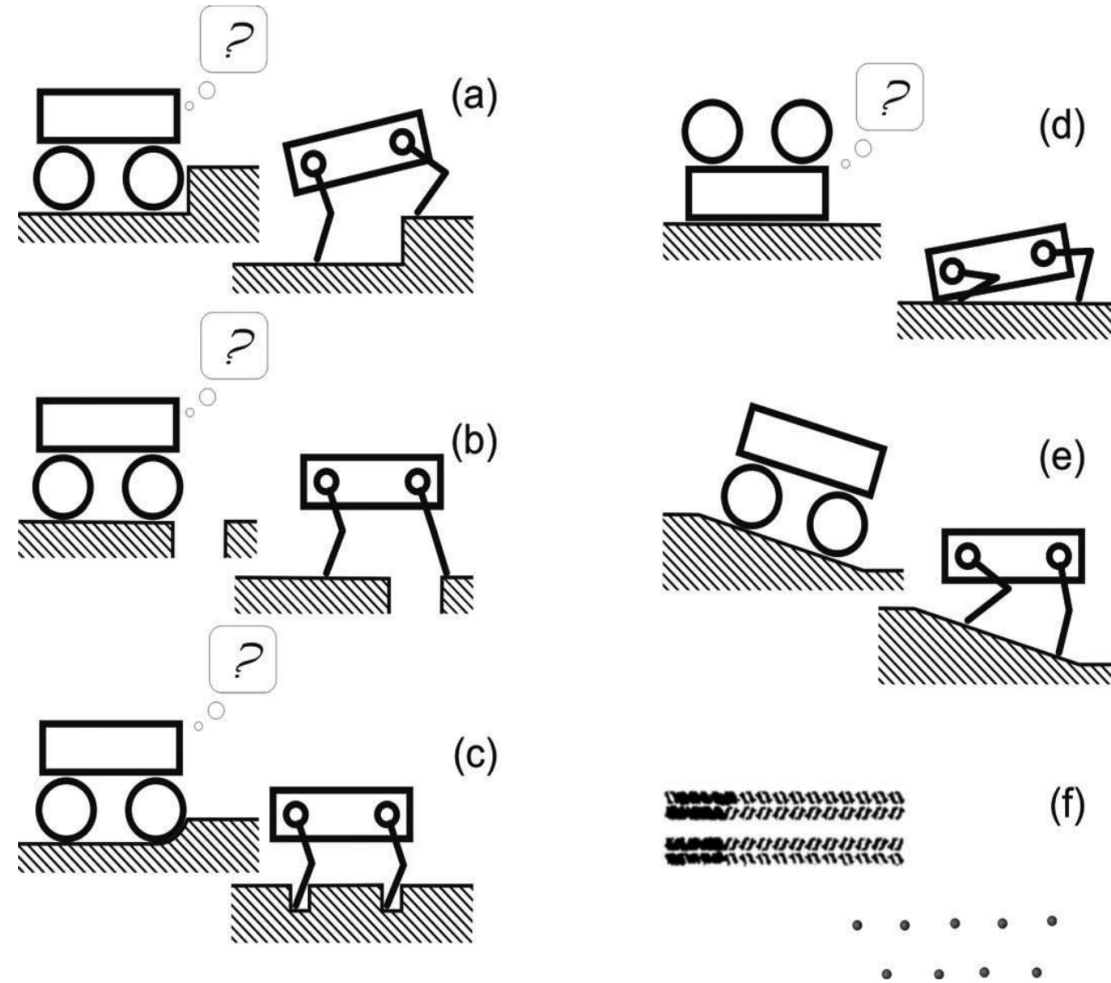


Walking Robots



Why legged robots?

- Adaptability and maneuverability in rough terrain. Legged systems can overcome many obstacles, cross a hole or chasm, that are not reachable by wheeled systems!
- The potential to manipulate objects in the environment with great skill.
- **But** it is quite hard to achieve this since power and mechanical complexity.
 - many DOFs must be controlled in a coordinated way.
 - the robot must see detailed elements of the terrain.



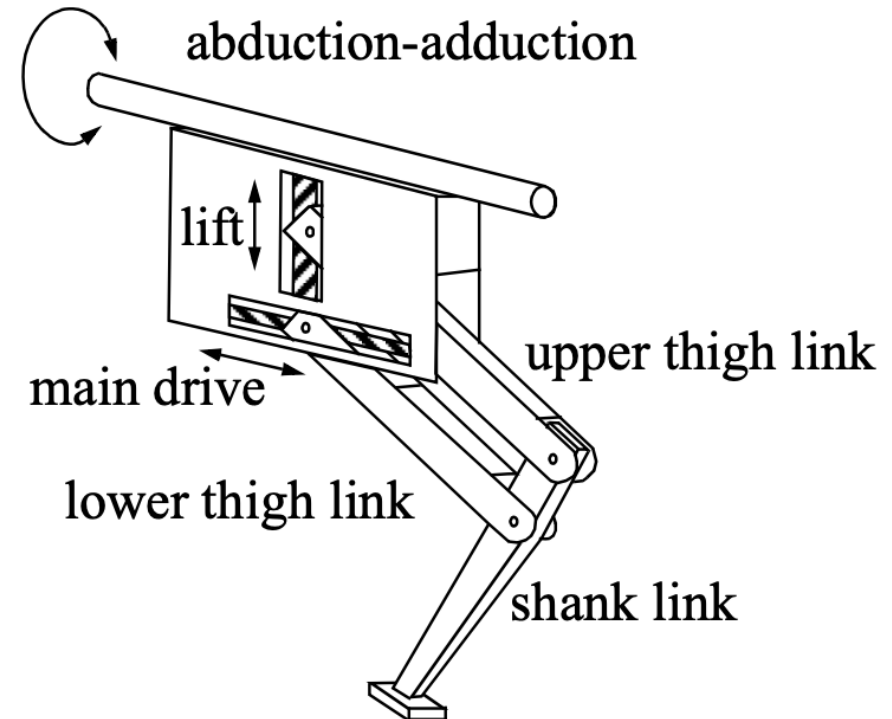
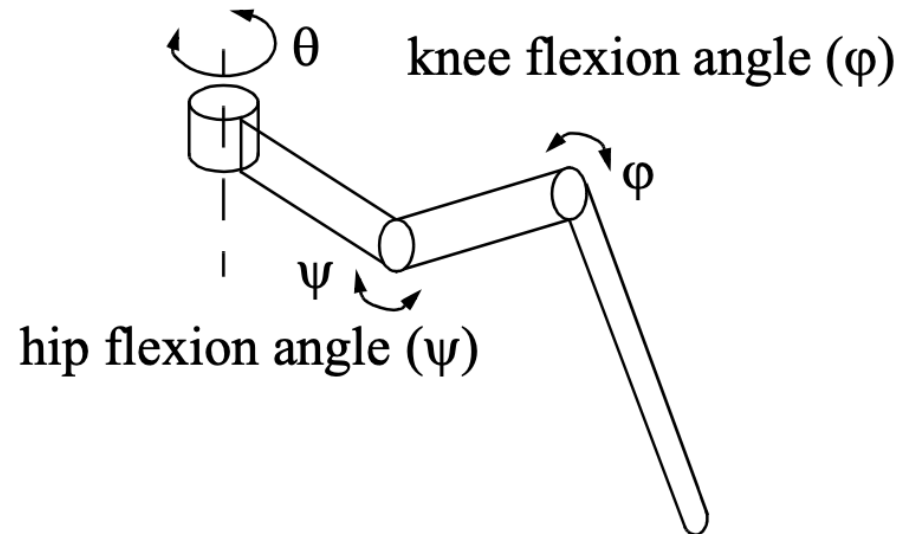
Number of Joints of Each Leg

- A minimum of **two** DOF is required to move a leg forward : a *lift* and a *swing* motion.
- The human leg has **more than seven** major DOF, combined with further actuation at the toes.
- Additional joint (DOF) increases the complexity of the design and especially of the locomotion control and maneuverability.
- The disadvantages of additional joints and actuators are energy, control, and mass.

Number of Joints of Each Leg

- Three DOF for each leg in most cases (e.g., pictured left below).
- 4th DOF for the ankle joint (e.g., pictured right below), might improve walking and stability.

hip abduction angle (θ)



The number of distinct event sequences (gaits)

- The gait is characterized as the distinct sequence of **lift and release events** of the individual legs

- It depends on the number of legs.

- The number of possible events N for a walking robot with k legs is:

$$N = (2k - 1)! \quad \text{factorial}$$

- The number 2 is associated with the number of events of a single leg.

- For a biped walker ($k=2$) the number of possible events N is:

$$N = (2k - 1)! = 3! = 3 * 2 * 1 = 6$$

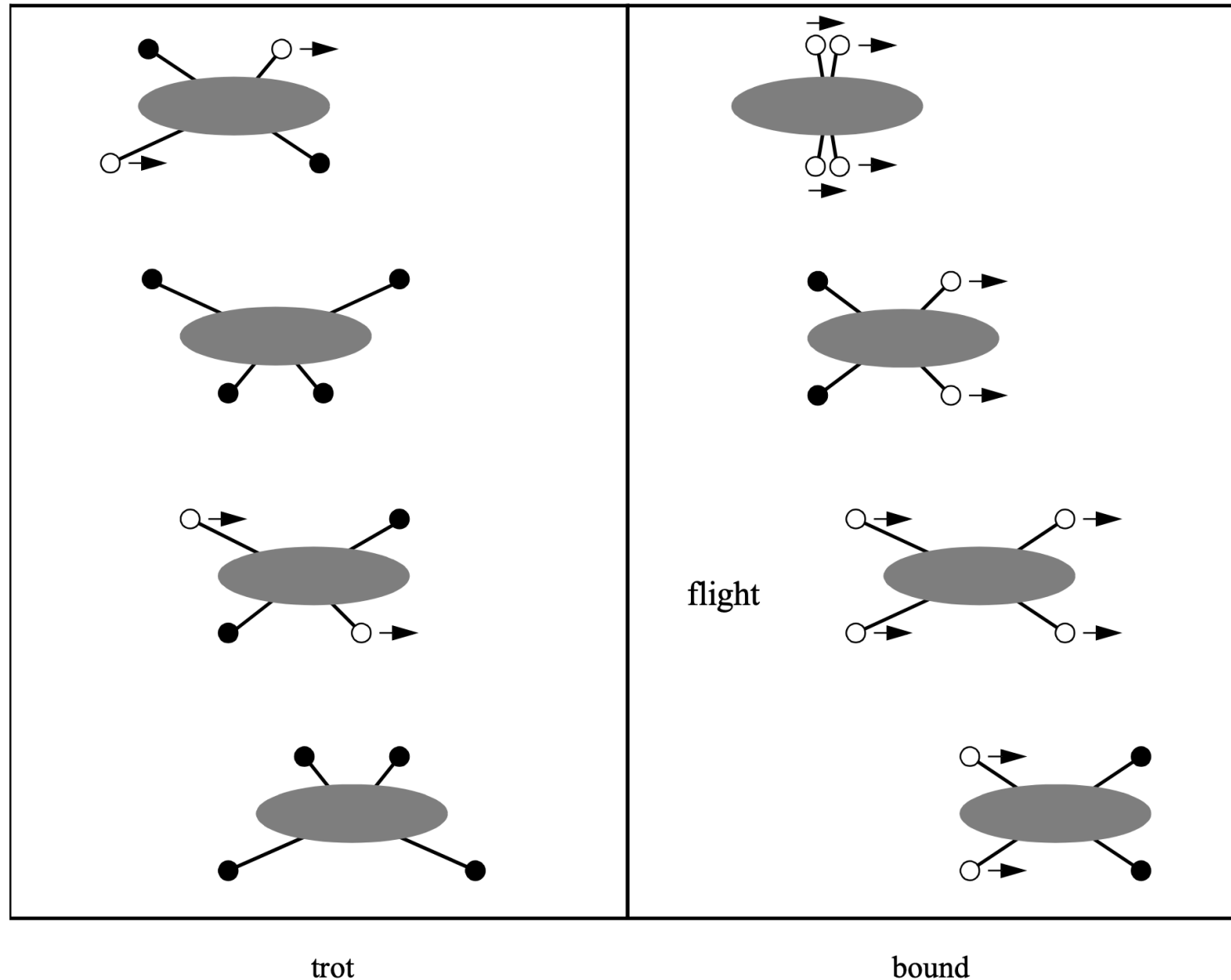
- For a robot with 6 legs (hexapod) N is already

$$N = 11! = 39'916'800$$

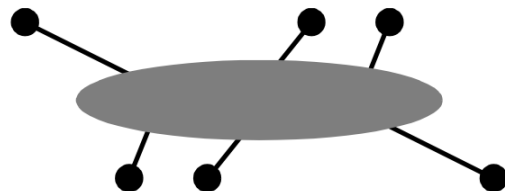
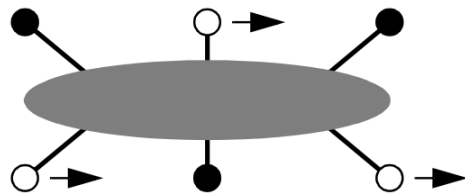
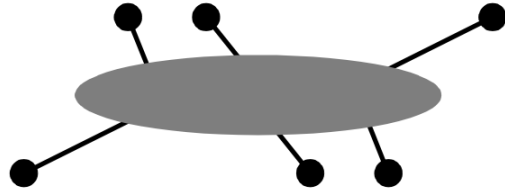
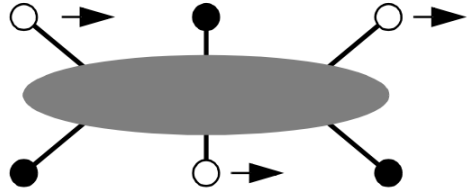
The number of distinct event sequences (gaits)

- For a biped walker ($k=2$) the number of possible events N is 6 as discussed previously.
 1. Both legs down – right down / left up – both legs down;
 2. Both legs down – right leg up / left leg down – both legs down;
 3. Right legs down – both legs up – both legs down;
 4. Right leg down / left leg up – right leg up / left leg down – right leg down / left leg up;
 5. Right leg down / left leg up – both legs up – right leg down / left leg up;
 6. Right leg up / left leg down – both legs up – right leg up / left leg down.

Most Obvious Natural Gaits with 4 Legs are Dynamic



Most Obvious Gaits with 6 Legs is Static



Static walking with six legs. A tripod formed by three legs always exists.

Legged Robotics: Impressive Mobility

- Spot – The Quadruped Robot (Boston Dynamics)



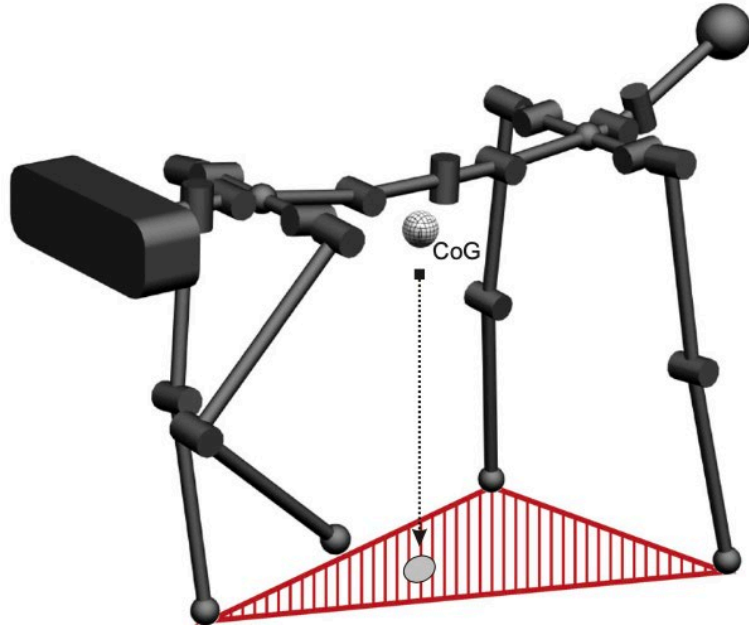
- Highly integrated leg design
- Large range of leg motion allowing for complex maneuvers
- "Static" walking: balance is maintained with no need for motion.

<https://www.bostondynamics.com/products>

<https://www.youtube.com/watch?v=zldyhGyXcUg>

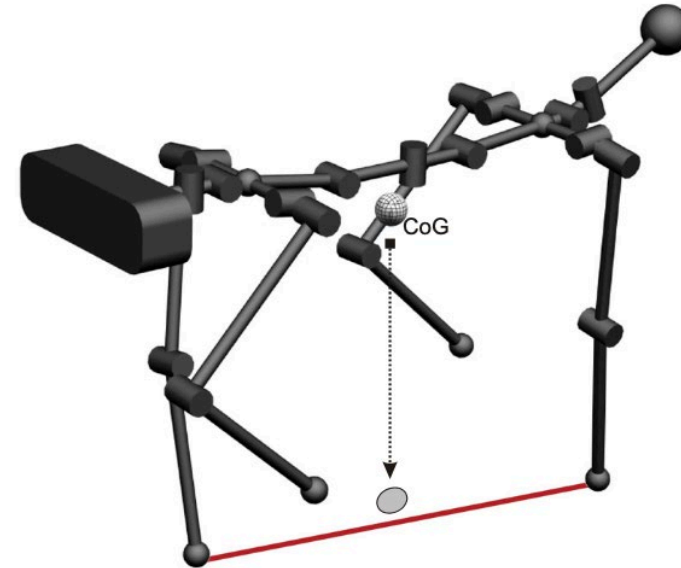
Dynamic vs. Static Walking

- Statically stable



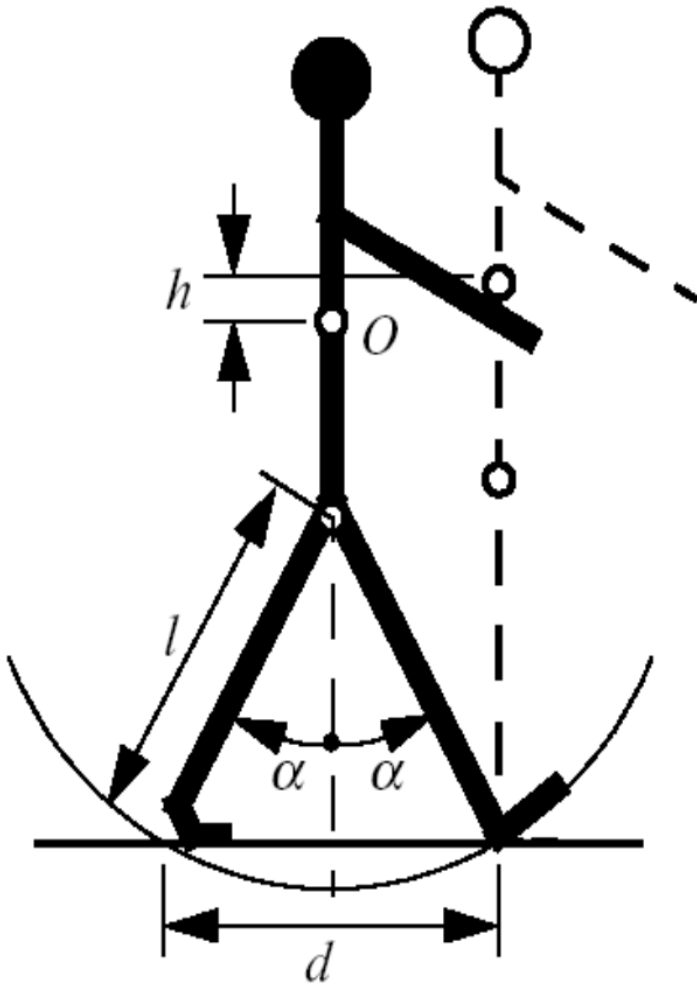
- Body weight supported by at least three legs
- Even if all joints 'freeze' instantaneously, the robot will not fall
- Safe, slow and inefficient

- Dynamic walking



- The robot will fall if not continuously moving
- Less than three legs can be in ground contact
- Fast, efficient and demanding for actuation and control

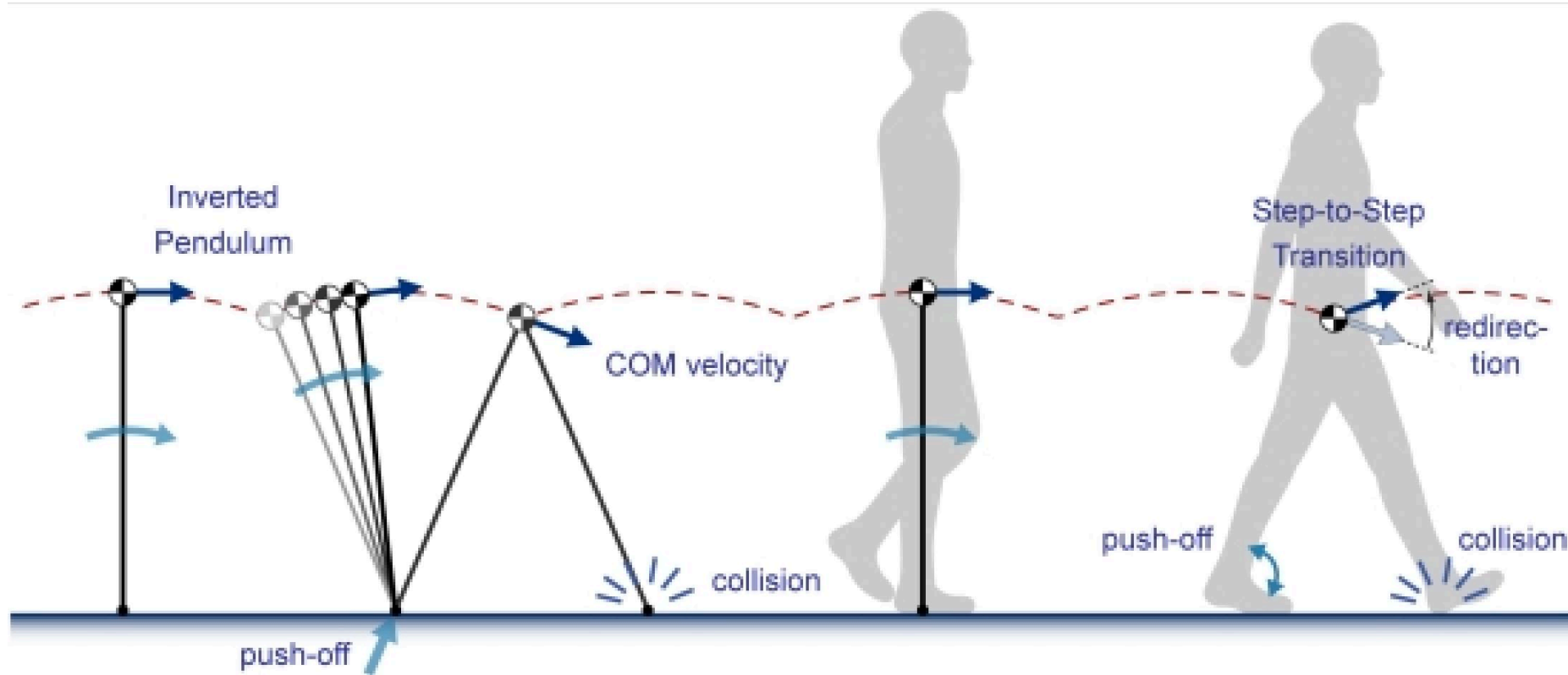
Biped Walking



- Biped walking mechanism
 - Not too far from real rolling when moving on flat ground
 - Rolling of a polygon with side length equal to the length of the step
 - The smaller the step gets, the more the polygon tends to a circle (wheel)
- But ...
 - Rotating joint was not invented by nature
 - Work against gravity is required

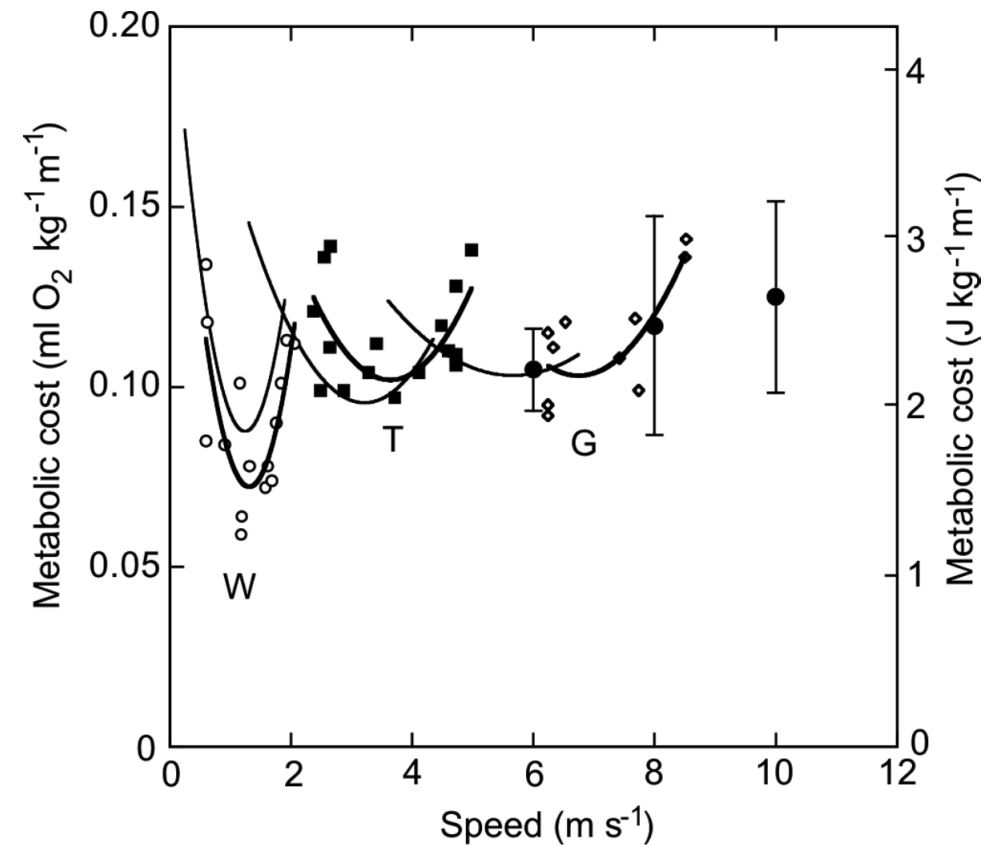
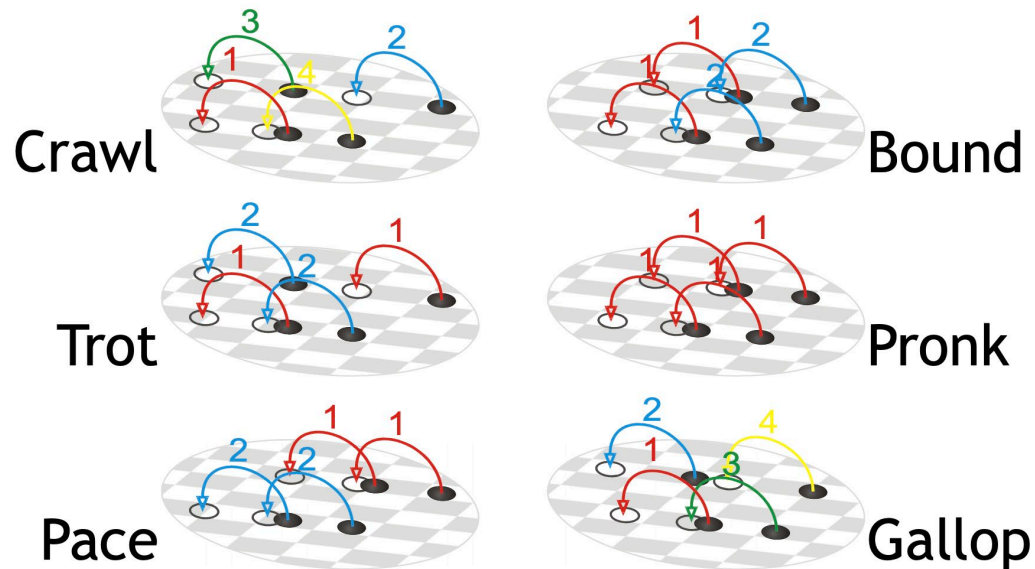
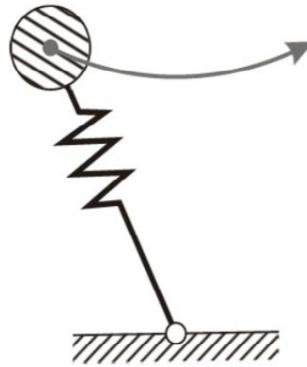
Modeling Legged Locomotion

- Inverted Pendulum
 - Walking can be represented by an inverted pendulum



Towards Efficient Dynamic Walking: Optimizing Gaits

- Nature optimizes its gaits
- Storage of “elastic” energy
- To allow locomotion at varying frequencies and speeds, different gaits must utilize these elements differently

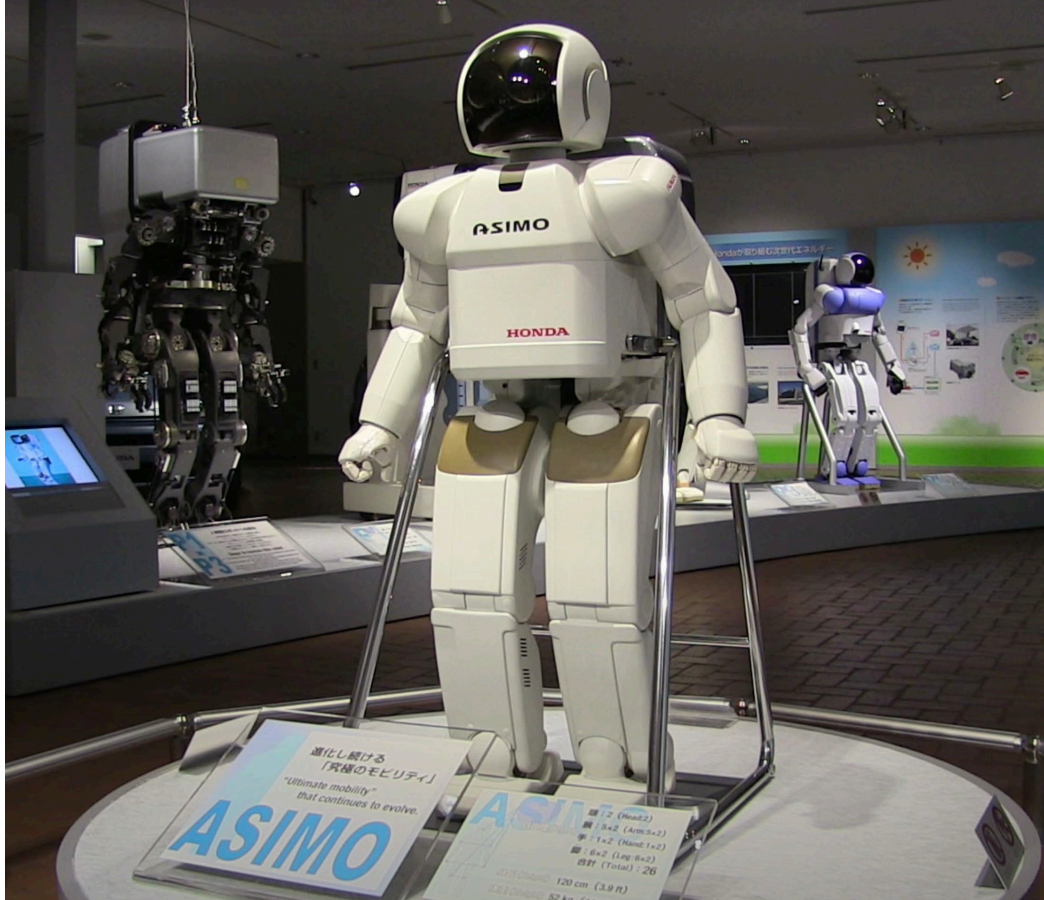


- The energetically most economic gait is a function of desired speed.

History of Legged Robotics

- Humanoid Robots after 2000

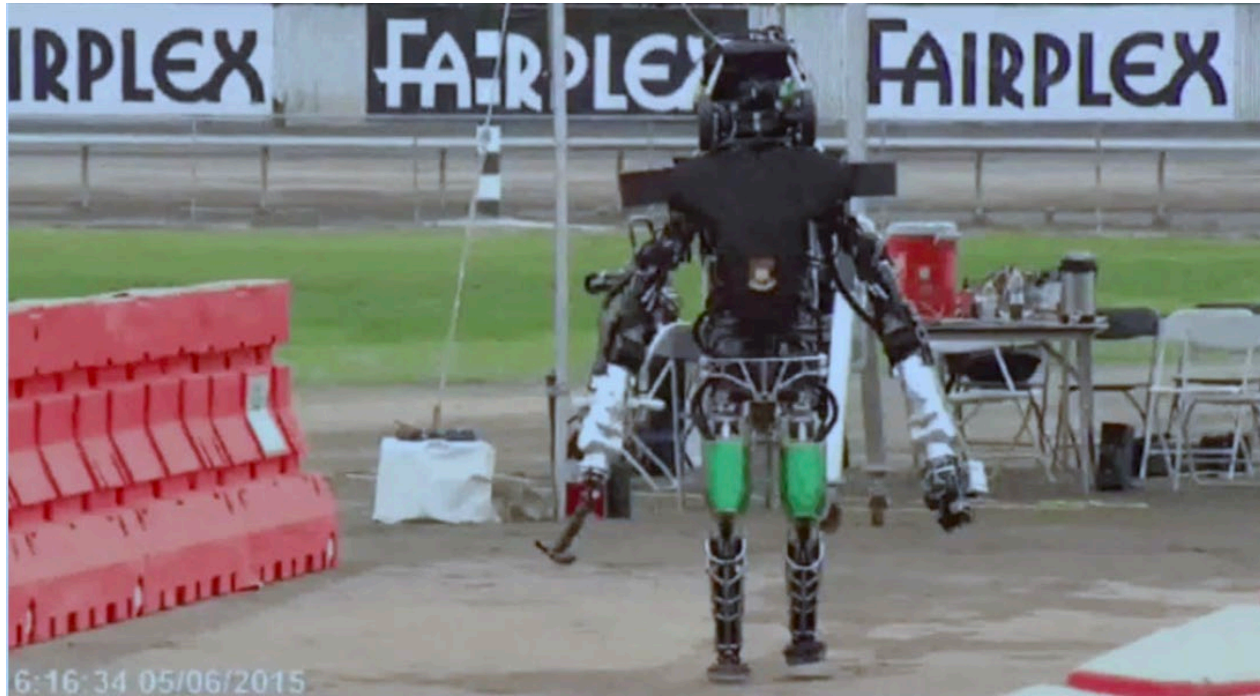
HONDA ASIMO



<https://www.youtube.com/watch?v=sIIUMLC3XfU>

History of Legged Robotics

- DARPA Robotics Challenge 2015



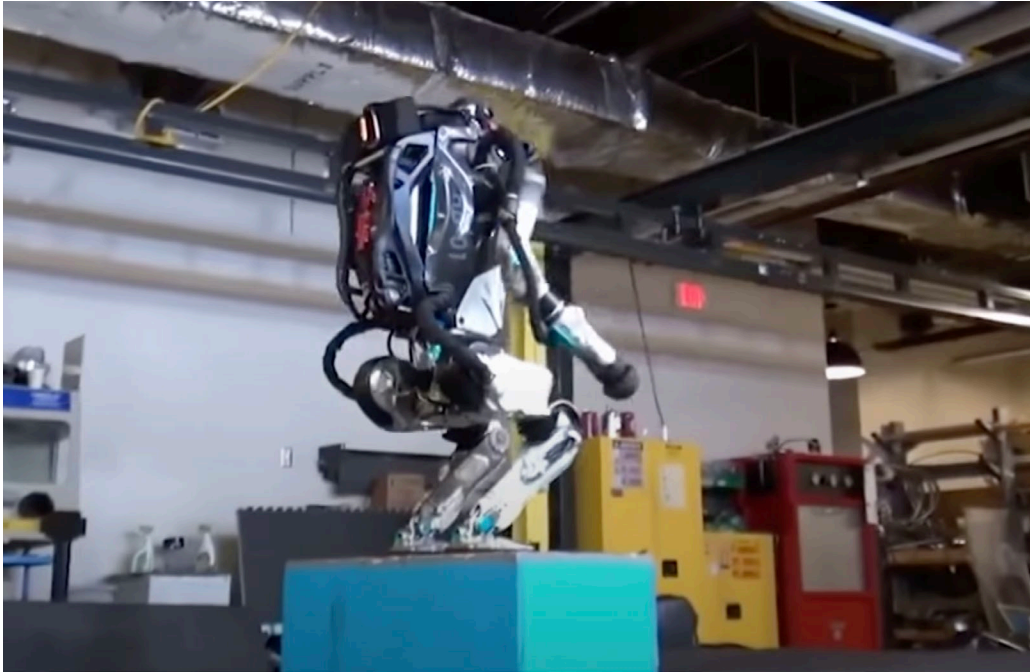
- A thing we learned is that walking is still difficult.



<https://www.youtube.com/watch?v=g0TaYhjpOfo>

History of Legged Robotics

- Boston Dynamics' Atlas robot: Impressive Mobility



<https://www.youtube.com/watch?v=knoOXBLFQ-s>

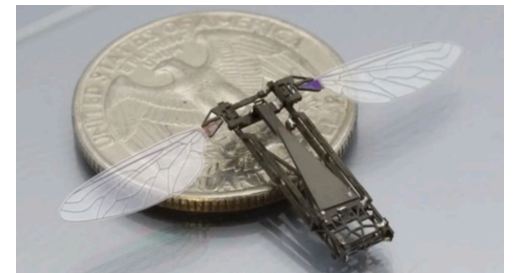
<https://www.youtube.com/watch?v=YlhzUnvi7Fw>

Flying Robots



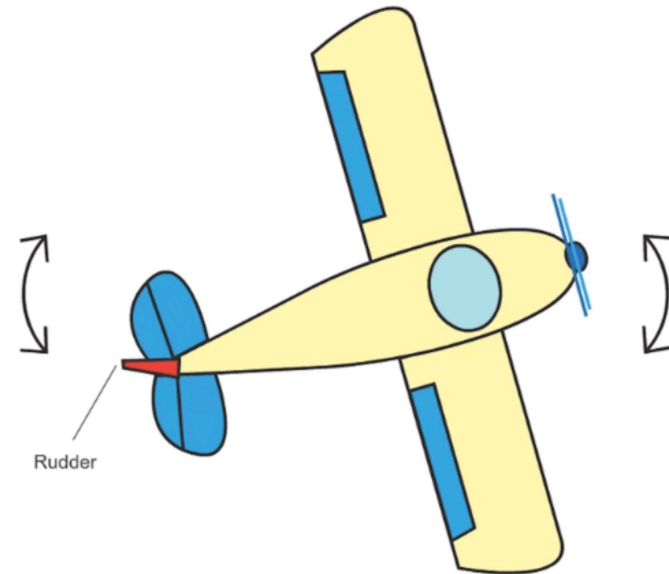
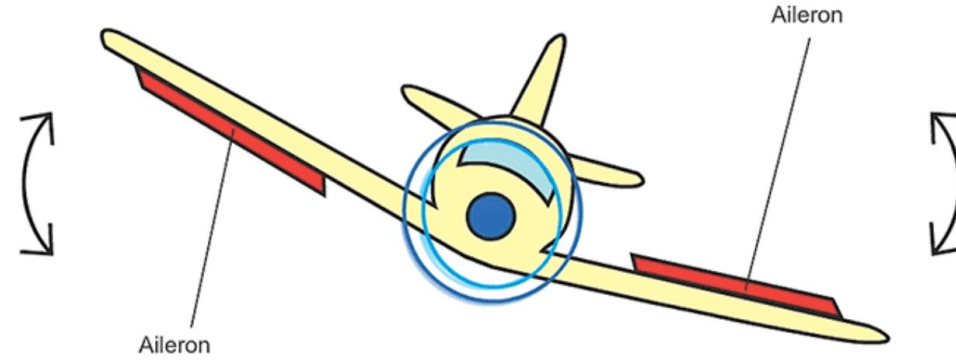
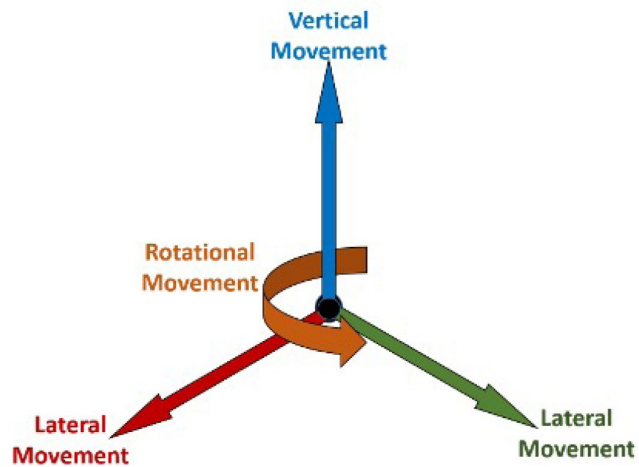
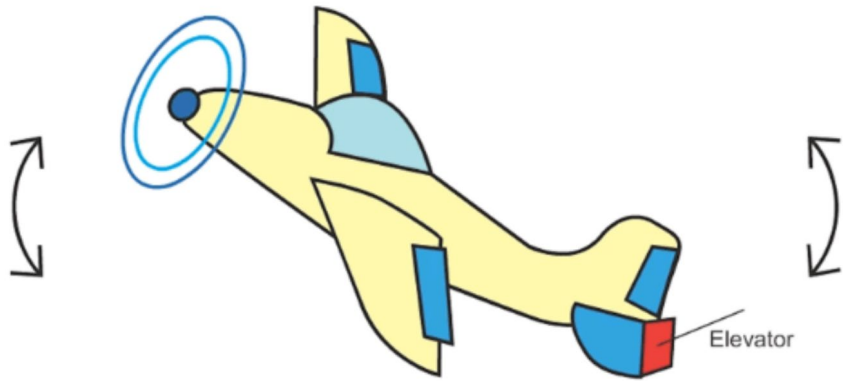
How to Fly: flight concepts (Unmanned Aerial Vehicles)

- Helicopters
 - < 20 minutes
 - Highly dynamic and agility
- Fixed Wing Airplanes
 - > some hours; continuous flights possible
 - Non-holonomic constraints
- Blimp: lighter-than-air
 - > some hours (dependent on wind conditions)
 - Sensitive to wind
 - Large size (dependent on payload)
- Flapping wings
 - < 20 minutes; gliding mode possible
 - Non-holonomic constraints
 - Very complex mechanics



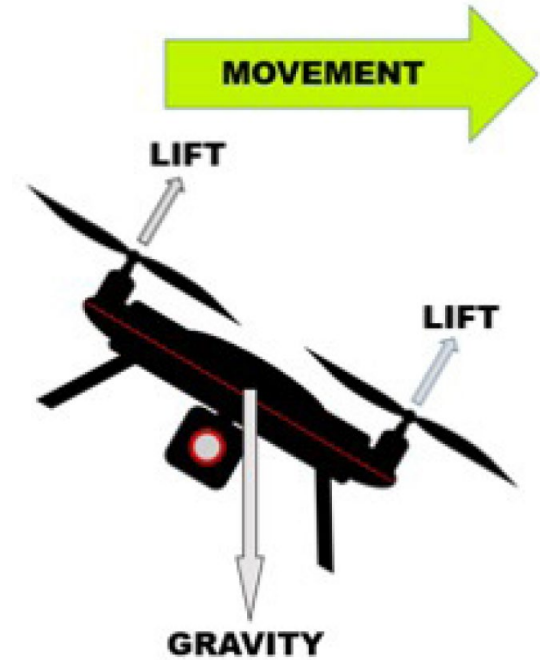
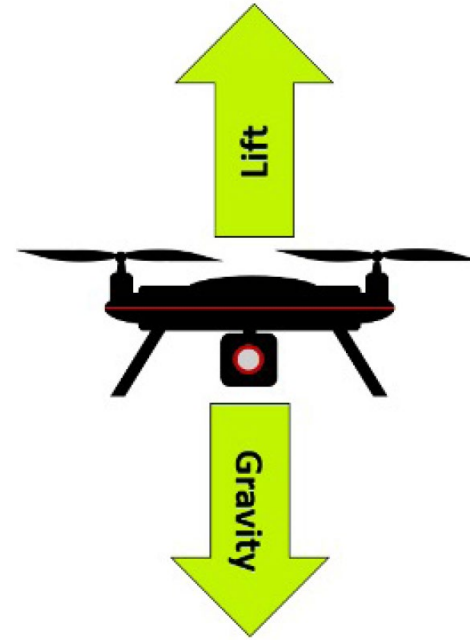
Movement of a Quadcopter

- Roll, Pitch, and Yaw



<https://howthingsfly.si.edu/flight-dynamics/roll-pitch-and-yaw>

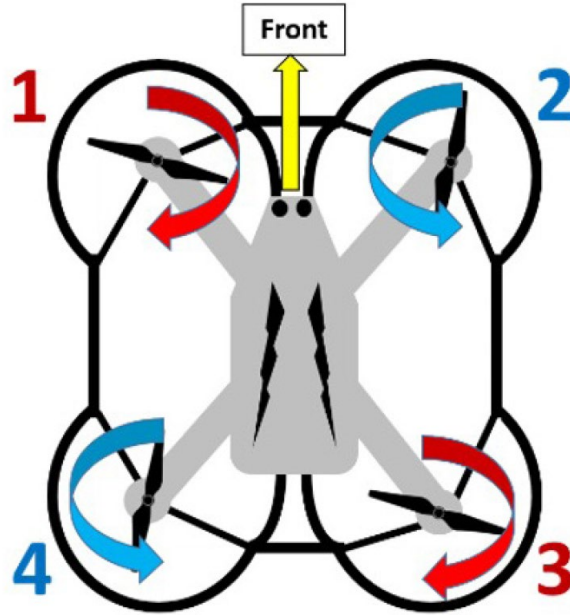
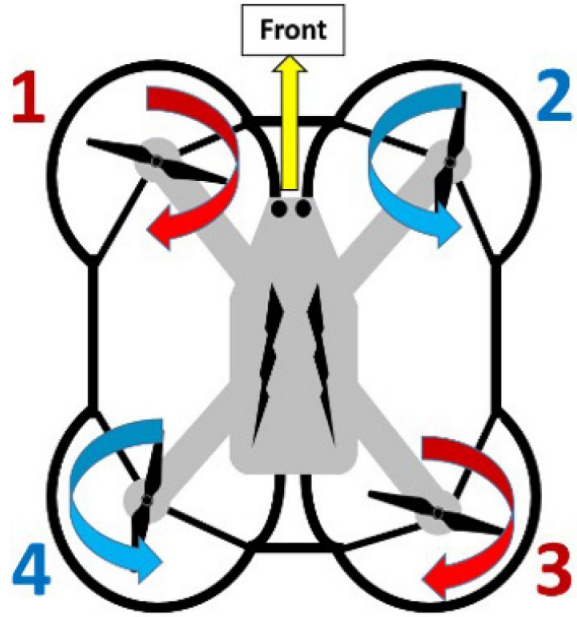
Movement of a Quadcopter



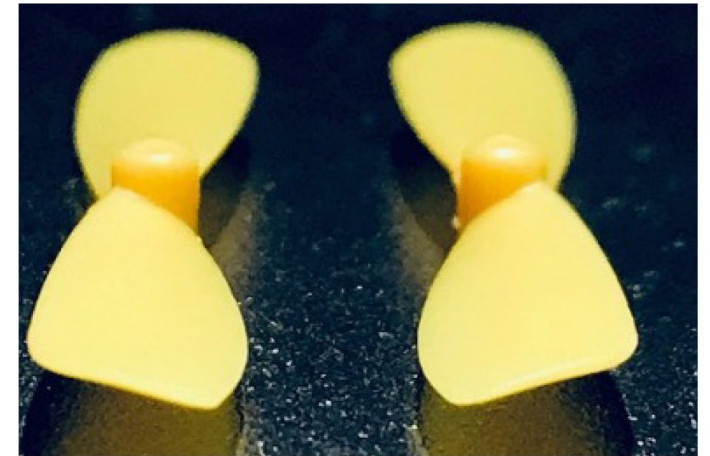
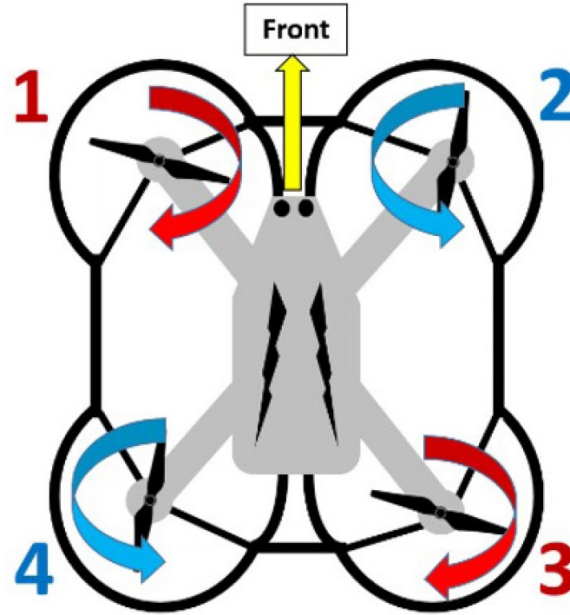
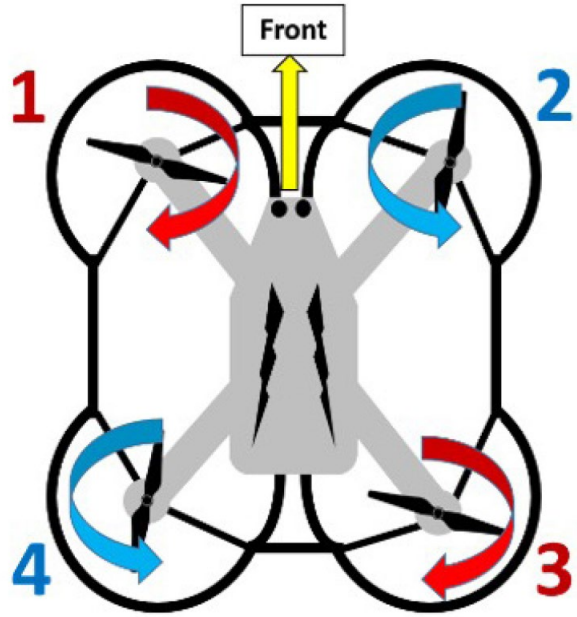
Movement of a Quadcopter



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Rules for Determining the Movement of the Quadcopter:

1. Rules for determining vertical movement: To calculate the total lift, add up the lifts created by each of the four propellers.
 - If the total lift is greater than the force of gravity (4 N), the quadcopter moves up.
 - If the total lift is equal to the force of gravity (4 N), the quadcopter doesn't move vertically.
 - If the total lift is less than the force of gravity (4 N), the quadcopter moves down.

Movement of a Quadcopter

2. Rules for determining lateral movement:

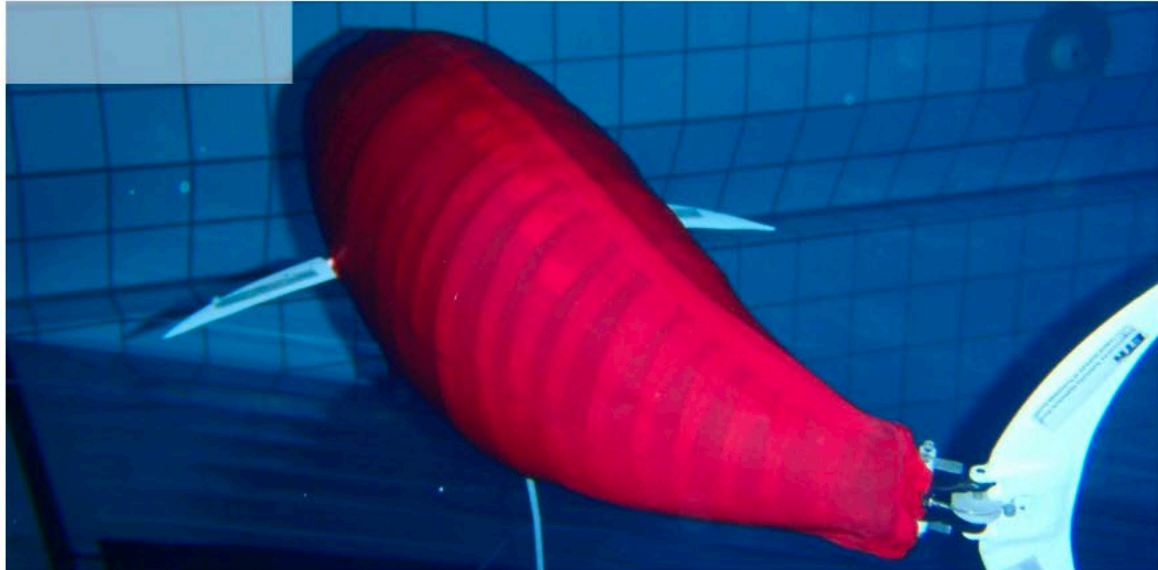
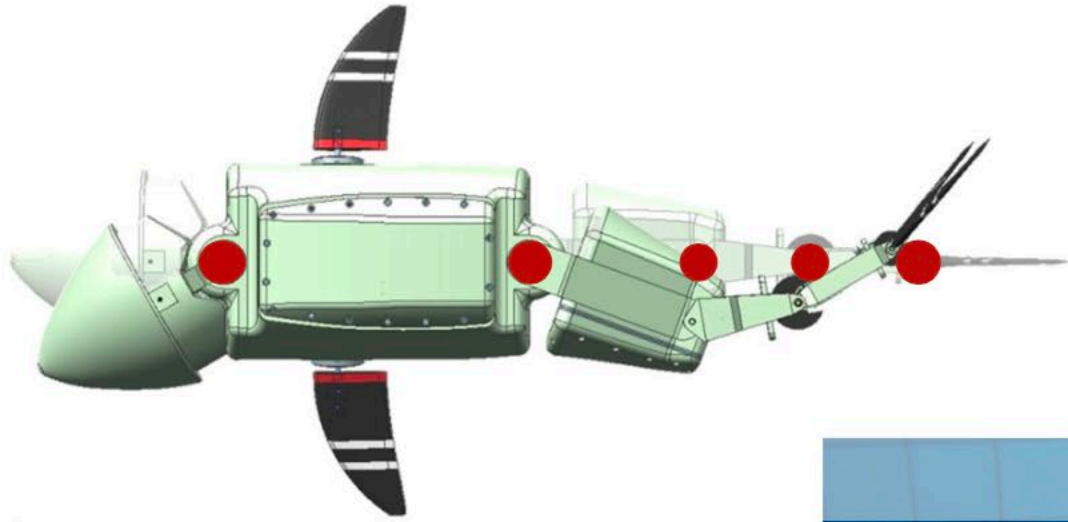
- If the lifts of propellers 1 and 4 add up to more than the lifts of propellers 2 and 3 added together, it moves right.
- If the lifts of propellers 2 and 3 add up to more than the lifts of propellers 1 and 4 added together, it moves left.
- If the lifts of propellers 1 and 2 add up to more than the lifts of propellers 3 and 4 added together, it moves backward.
- If the lifts of propellers 3 and 4 add up to more than the lifts of propellers 1 and 2 added together, it moves forward.

Movement of a Quadcopter

3. Rules for determining rotational movement:

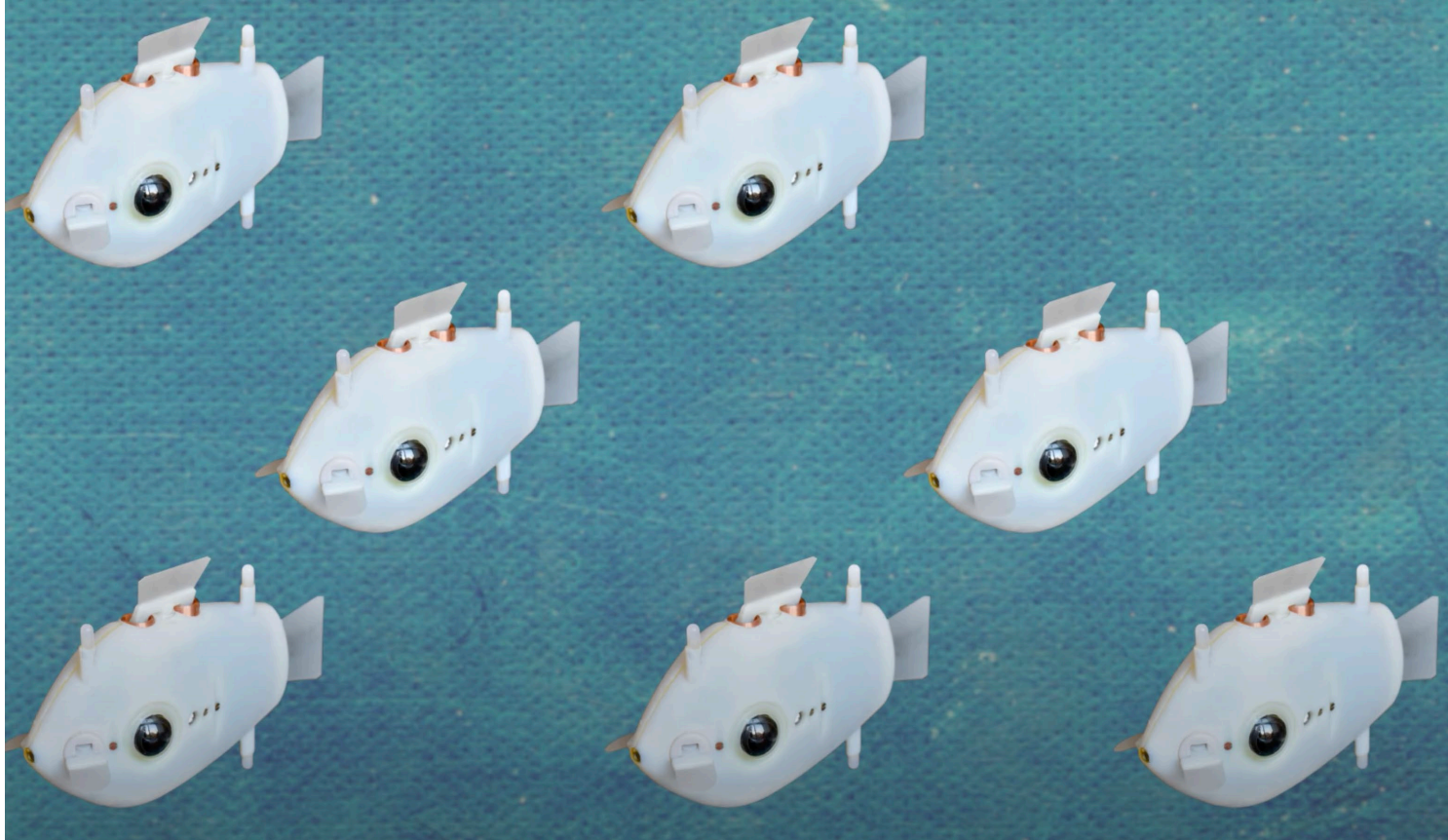
- If the lifts of propellers 1 and 3 add up to more than the lifts of propellers 2 and 4, it rotates counterclockwise.
- If the lifts of propellers 2 and 4 add up to more than the lifts of propellers 1 and 3, it rotates clockwise.
- Otherwise, there is no rotational movement.

Swimming Robots



Underwater Robots

- An example of swarms of robotic fish



<https://www.youtube.com/watch?v=qVsu49f-Vf0>